

# Topic 3: Fourier Transform properties

Linearity, shift, symmetry, convolution, windows, FFT and analysis/synthesis

Digital Signal Processing for Sound and Music · Topic 3

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# Lecture roadmap

- 1 Linearity, shift and symmetry
- 3 Energy, dB and phase unwrapping
- 5 FFT and analysis/synthesis workflow

- 2 Convolution and filtering intuition
- 4 Zero-padding and zero-phase FFT buffers
- 6 Example notebook + Exercise 3

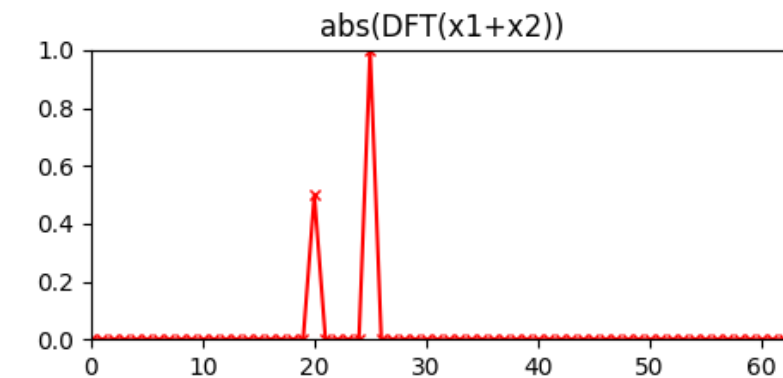
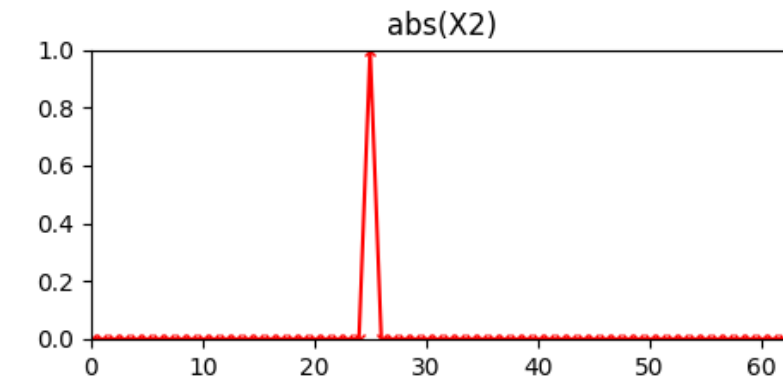
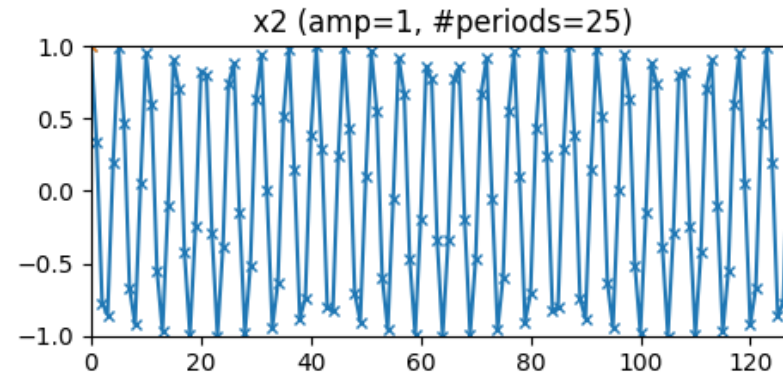
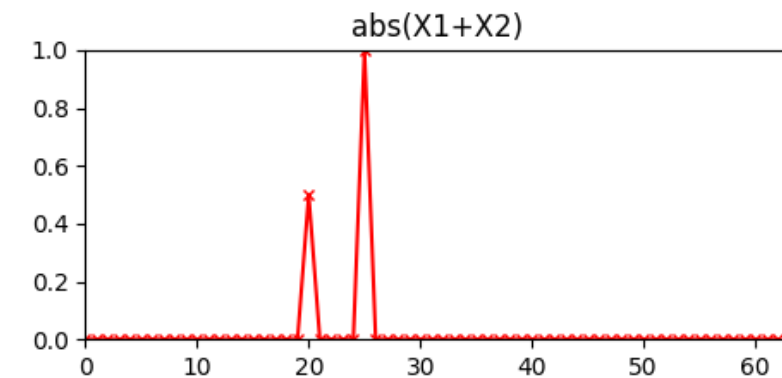
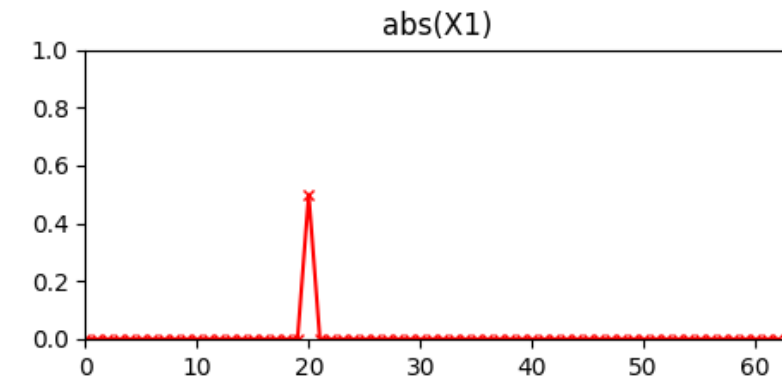
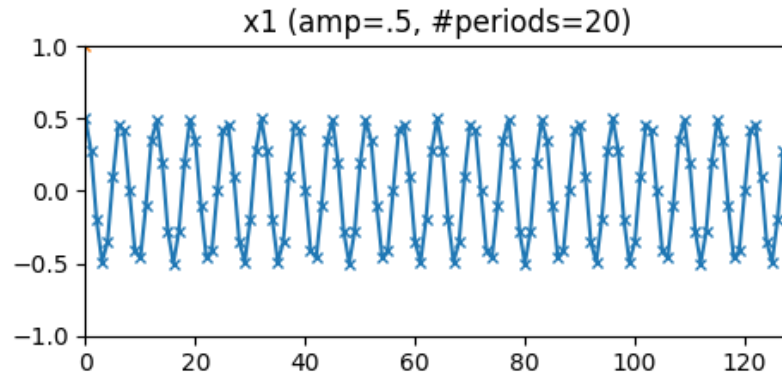
**Linearity:** Property:  $ax_1[n] + bx_2[n] \Leftrightarrow aX_1[k] + bX_2[k]$

$$\begin{aligned} & DFT(ax_1[n] + bx_2[n]) \\ &= \sum_{n=0}^{N-1} (ax_1[n] + bx_2[n])e^{-j2\pi kn/N} \\ &= \sum_{n=0}^{N-1} a x_1[n]e^{-j2\pi kn/N} + \sum_{n=0}^{N-1} b x_2[n]e^{-j2\pi kn/N} \\ &= a \sum_{n=0}^{N-1} x_1[n]e^{-j2\pi kn/N} + b \sum_{n=0}^{N-1} x_2[n]e^{-j2\pi kn/N} \\ &= aX_1[k] + bX_2[k] \end{aligned}$$

### Linearity

- DFT is a linear transform.
- Analyze sums by summing spectra.
- Useful for additive sound models.

# Linearity: spectra add



# Shift:

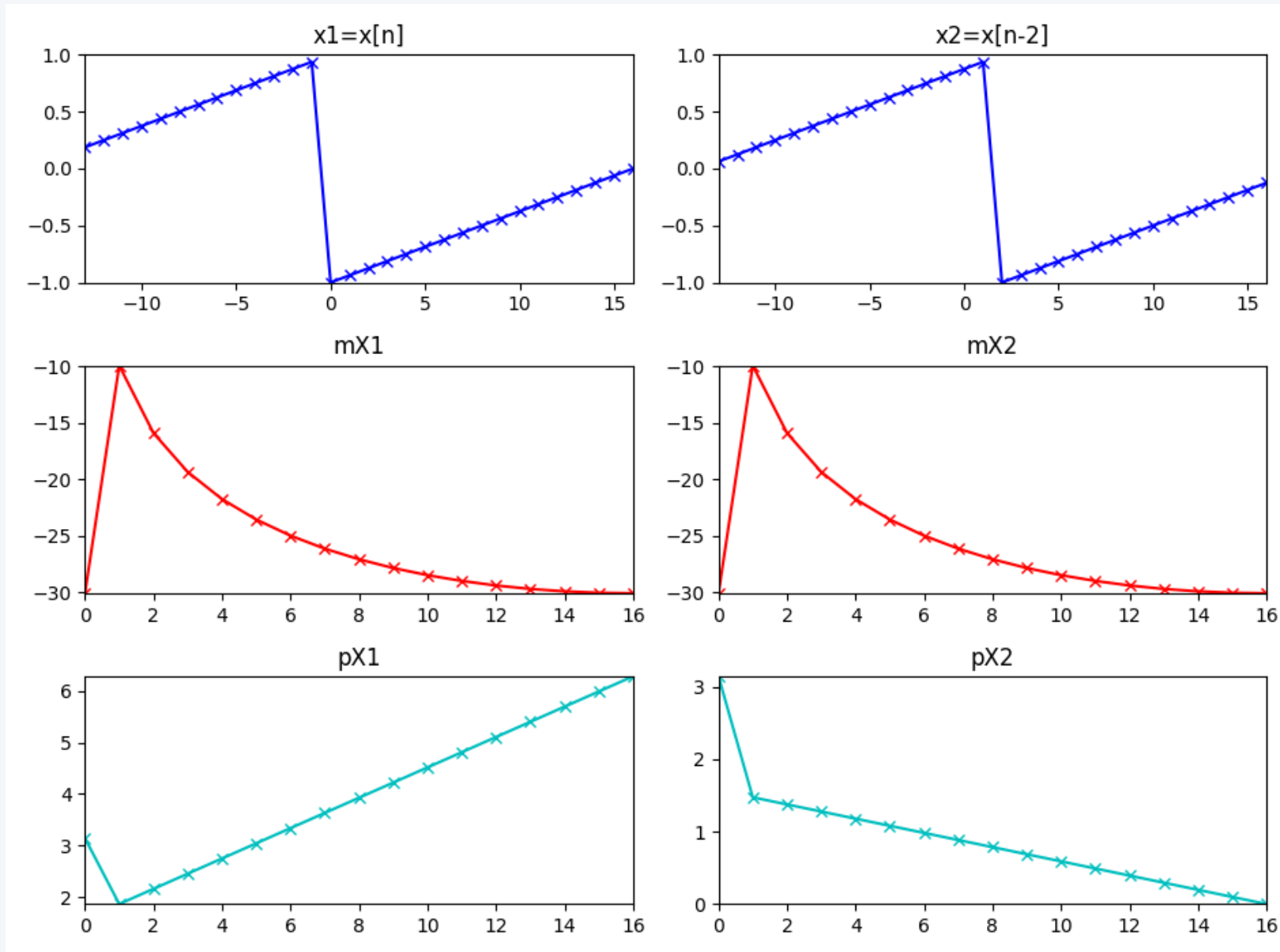
Property:  $x[n - n_0] \Leftrightarrow e^{-j2\pi kn_0/N} X[k]$

$$\begin{aligned} & DFT(x[n - n_0]) \\ &= \sum_{n=0}^{N-1} x[n - n_0] e^{-j2\pi kn/N} \\ &= \sum_{m=-n_0}^{N-1-n_0} x[m] e^{-j2\pi k(m+n_0)/N} \quad (m = n - n_0) \\ &= \sum_{m=0}^{N-1} x[m] e^{-j2\pi km/N} e^{-j2\pi kn_0/N} \\ &= e^{-j2\pi kn_0/N} \sum_{m=0}^{N-1} x[m] e^{-j2\pi km/N} \\ &= e^{-j2\pi kn_0/N} X[k] \end{aligned}$$

## Shift

- A time shift does not change magnitude.
- It adds a linear phase term.
- This is why centering matters.

# Shift: same magnitude, phase rotation

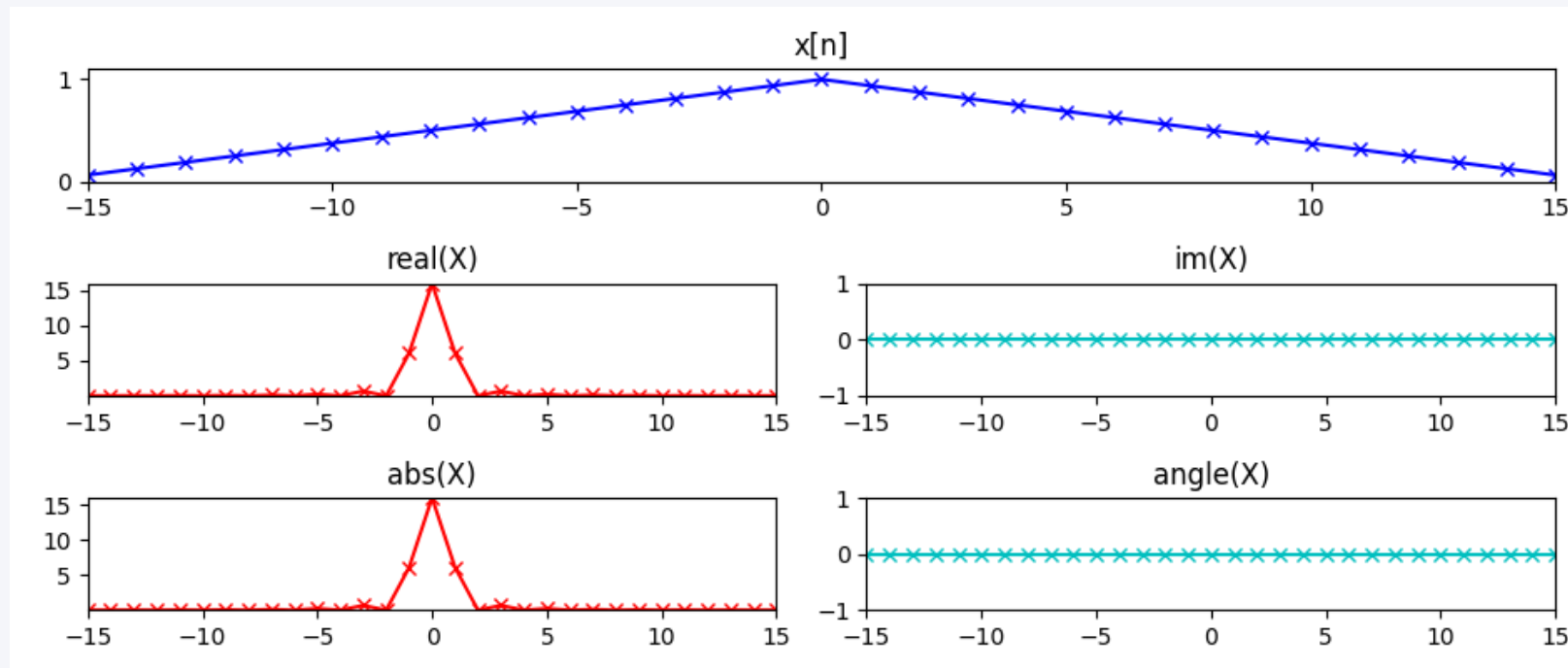


# Symmetry:

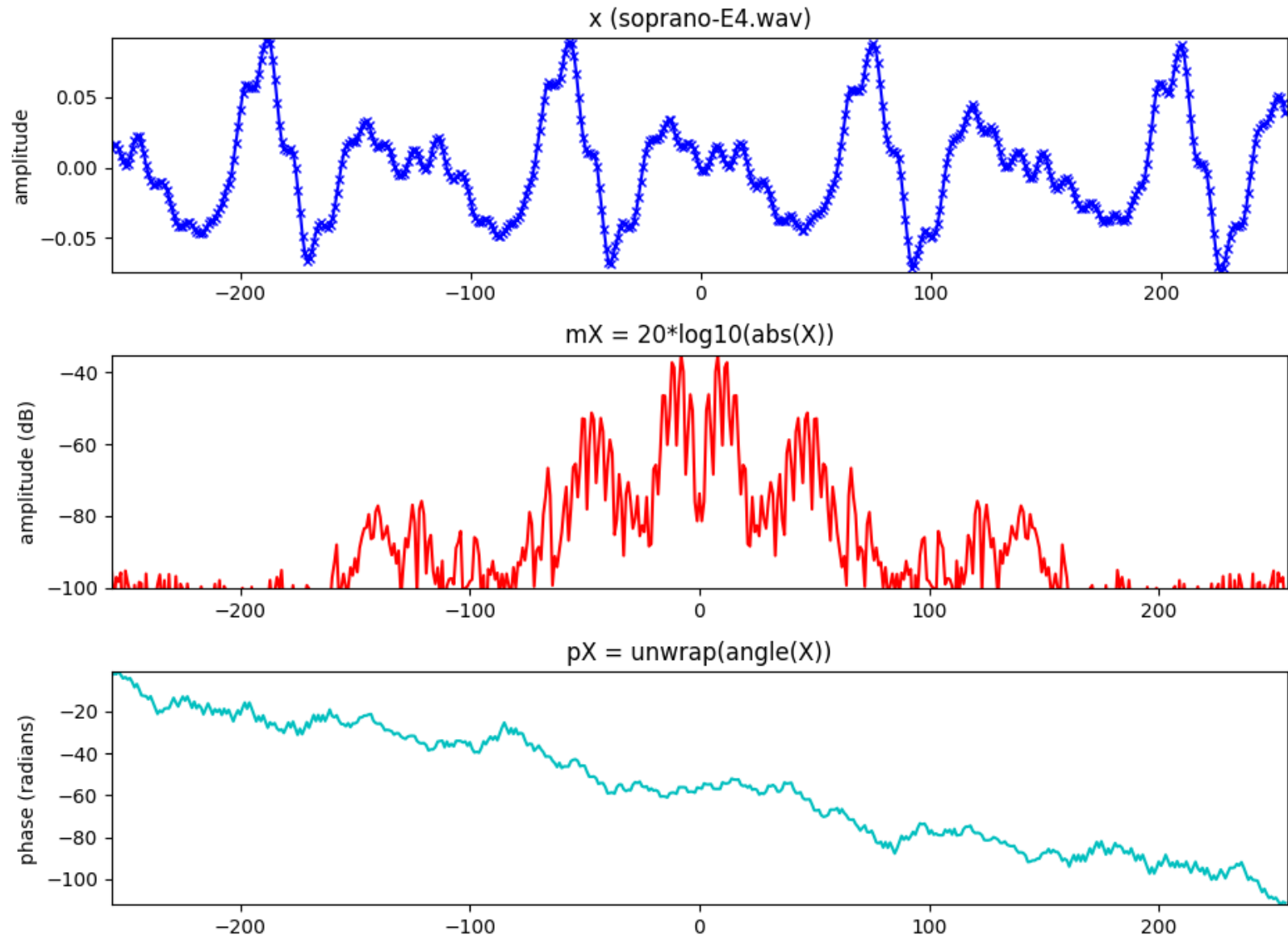
$$\begin{aligned}x[n] \text{ real} &\Leftrightarrow \Re\{X[k]\} \text{ even and } \Im\{X[k]\} \text{ odd} \\&\Leftrightarrow |X[k]| \text{ even and } \angle X[k] \text{ odd} \\x[n] \text{ real and even} &\Leftrightarrow \Re\{X[k]\} \text{ even and } \Im\{X[k]\} = 0 \\&\Leftrightarrow |X[k]| \text{ even and } \angle X[k] = n\pi\end{aligned}$$

## Symmetry

- Real signals have conjugate-symmetric spectra.
- Real + even signals have real spectra.
- These properties reduce storage and computation.



# Symmetry for a centered real signal



# Convolution:

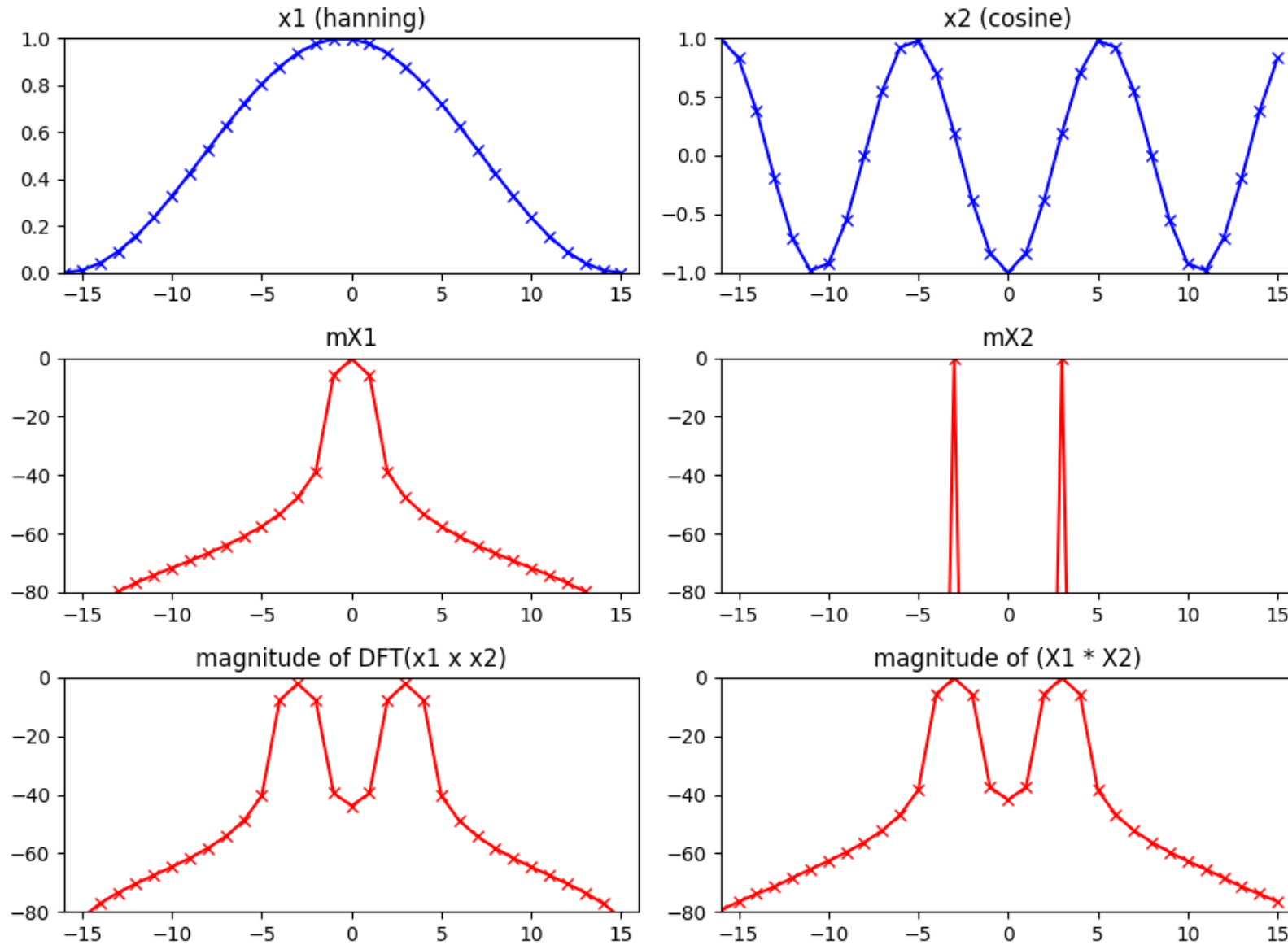
Property:  $x_1[n] * x_2[n] \Leftrightarrow X_1[k] \times X_2[k]$

$$\begin{aligned} & DFT(x_1[n] * x_2[n]) \\ &= \sum_{n=0}^{N-1} (x_1[n] * x_2[n]) e^{-j2\pi kn/N} \\ &= \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} x_1[m] x_2[n-m] e^{-j2\pi kn/N} \\ &= \sum_{m=0}^{N-1} x_1[m] \sum_{n=0}^{N-1} x_2[n-m] e^{-j2\pi kn/N} \\ &= \left( \sum_{m=0}^{N-1} x_1[m] e^{-j2\pi km/N} \right) X_2[k] \\ &= X_1[k] \times X_2[k] \end{aligned}$$

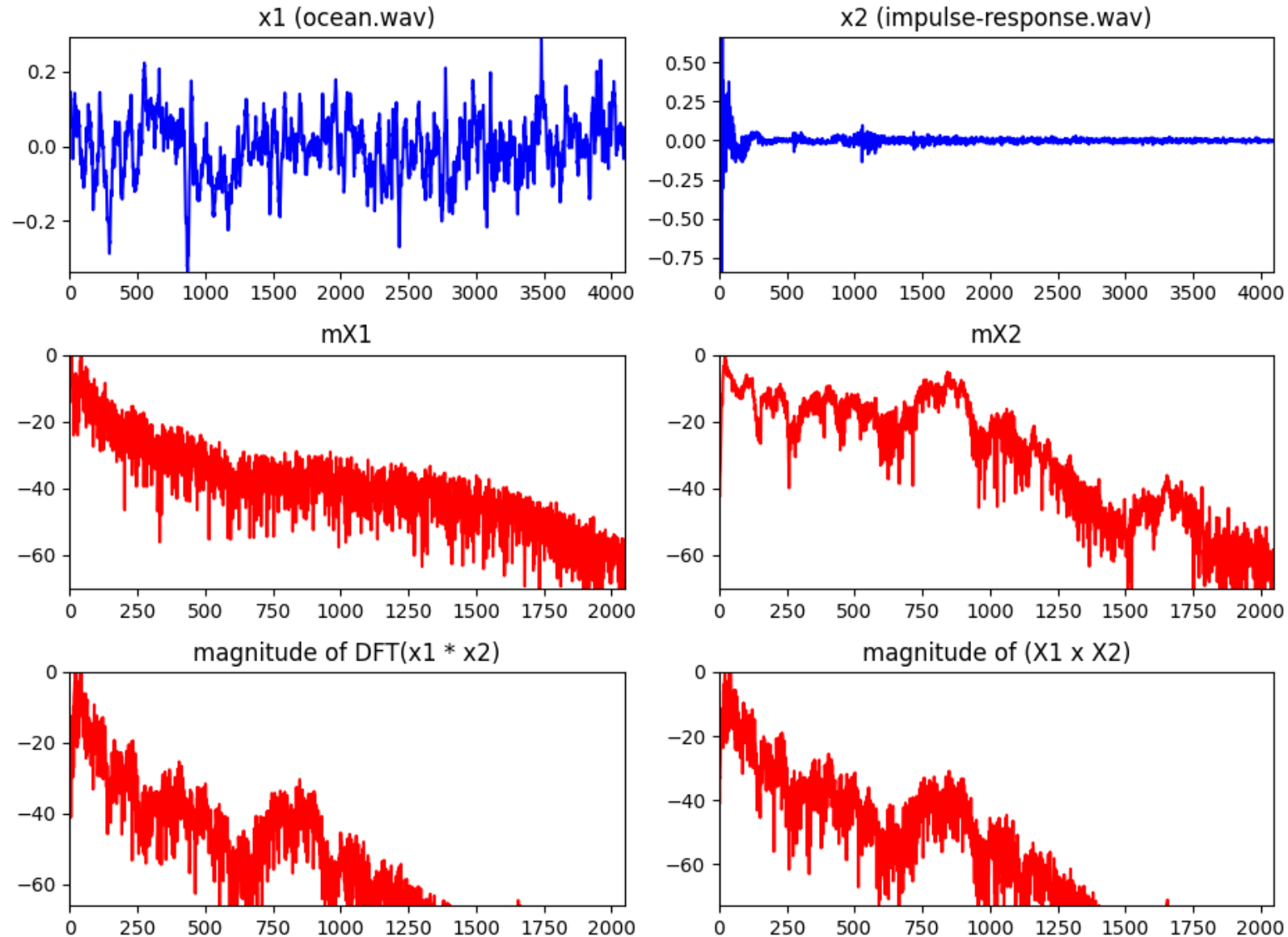
## Property:

- Convolution in time becomes multiplication in frequency.
- This is the basis of filtering.
- Circular convolution follows directly from the DFT.

# Convolution property



# Convolution as frequency-domain filtering



# Energy conservation

Parseval identity:

$$\sum |x[n]|^2 = (1/N) \sum |X[k]|^2$$

time energy:

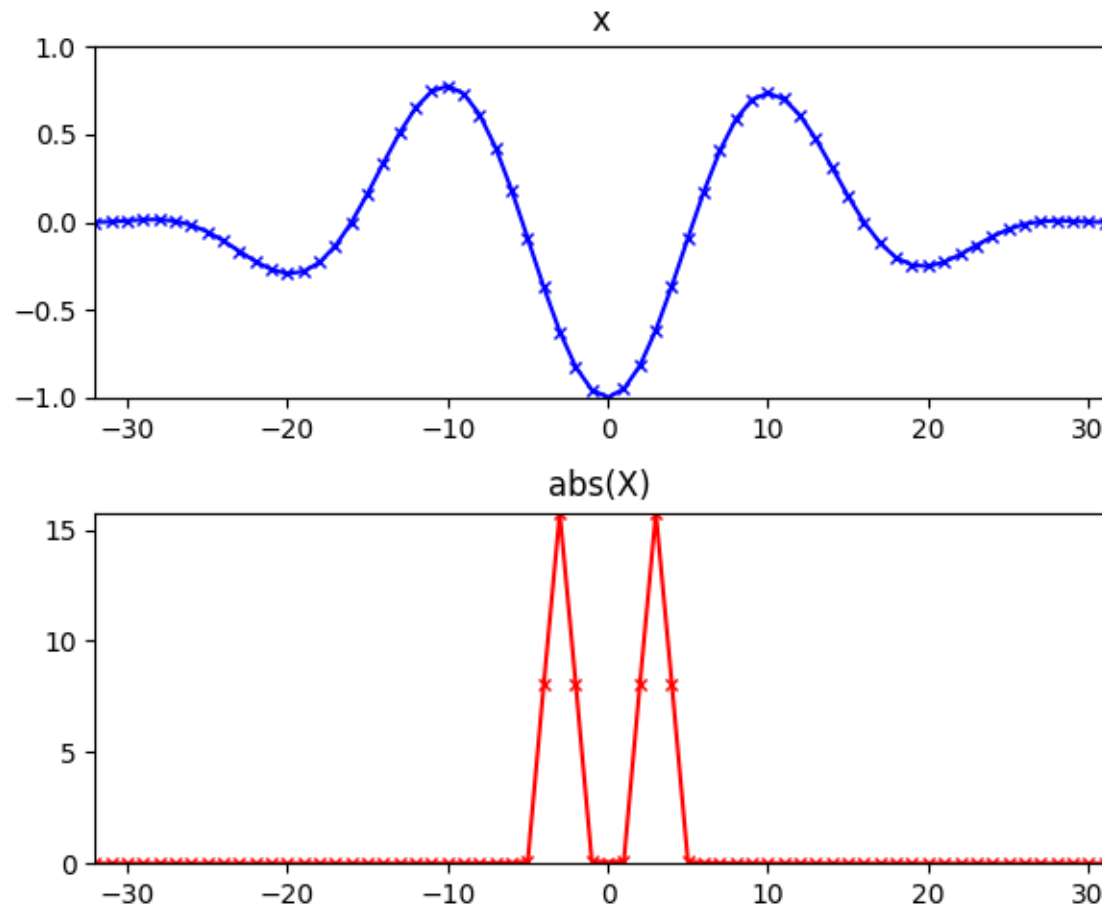
$$\sum |x[n]|^2 = 11.81182$$

frequency energy:

$$(1/N) \sum |X[k]|^2 = 11.81182$$

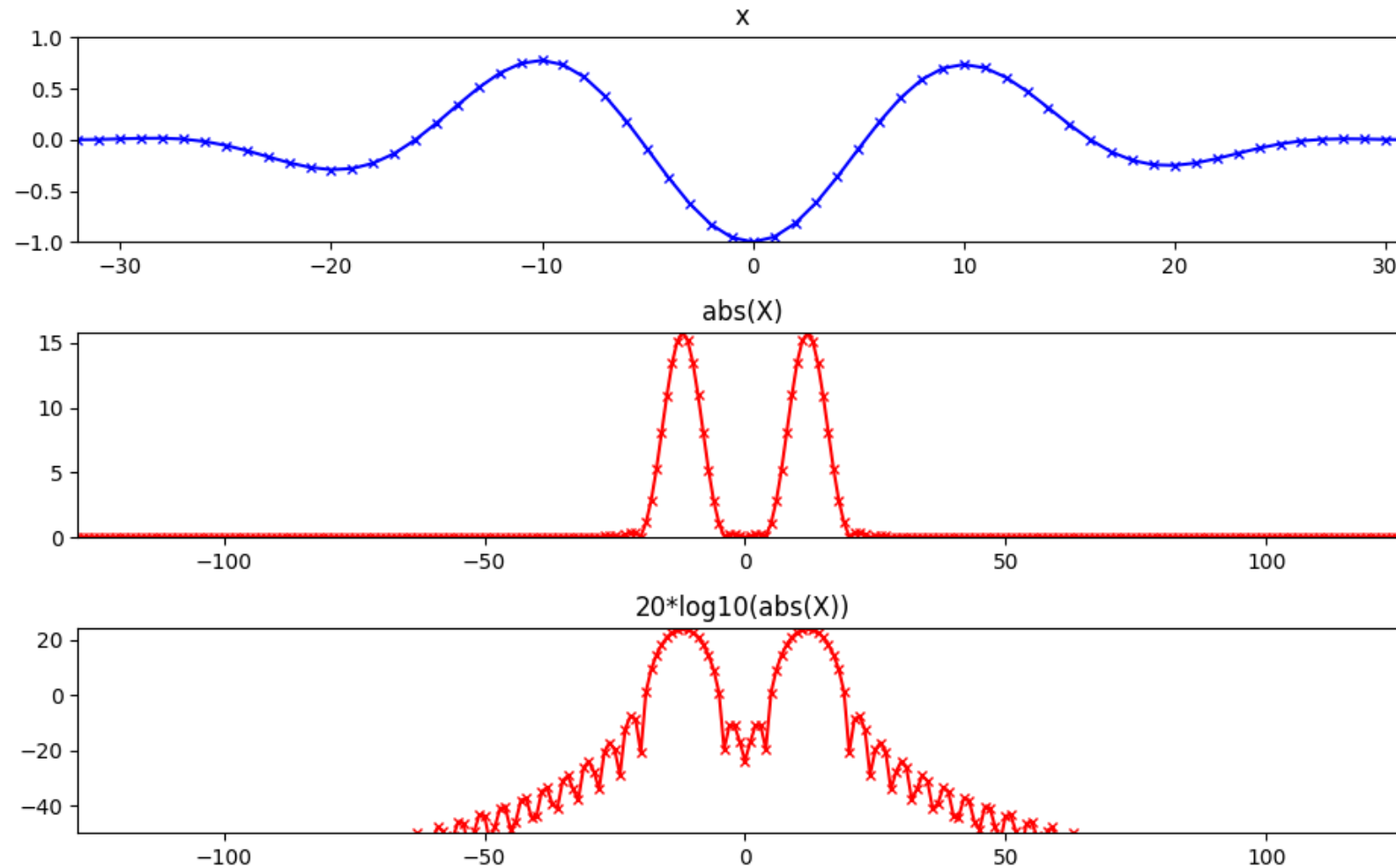
## Parseval

- Energy is conserved across domains.
- Frequency-domain energy has  $1/N$  scaling.
- Useful for checking implementations.



# Amplitude in decibels (dB)

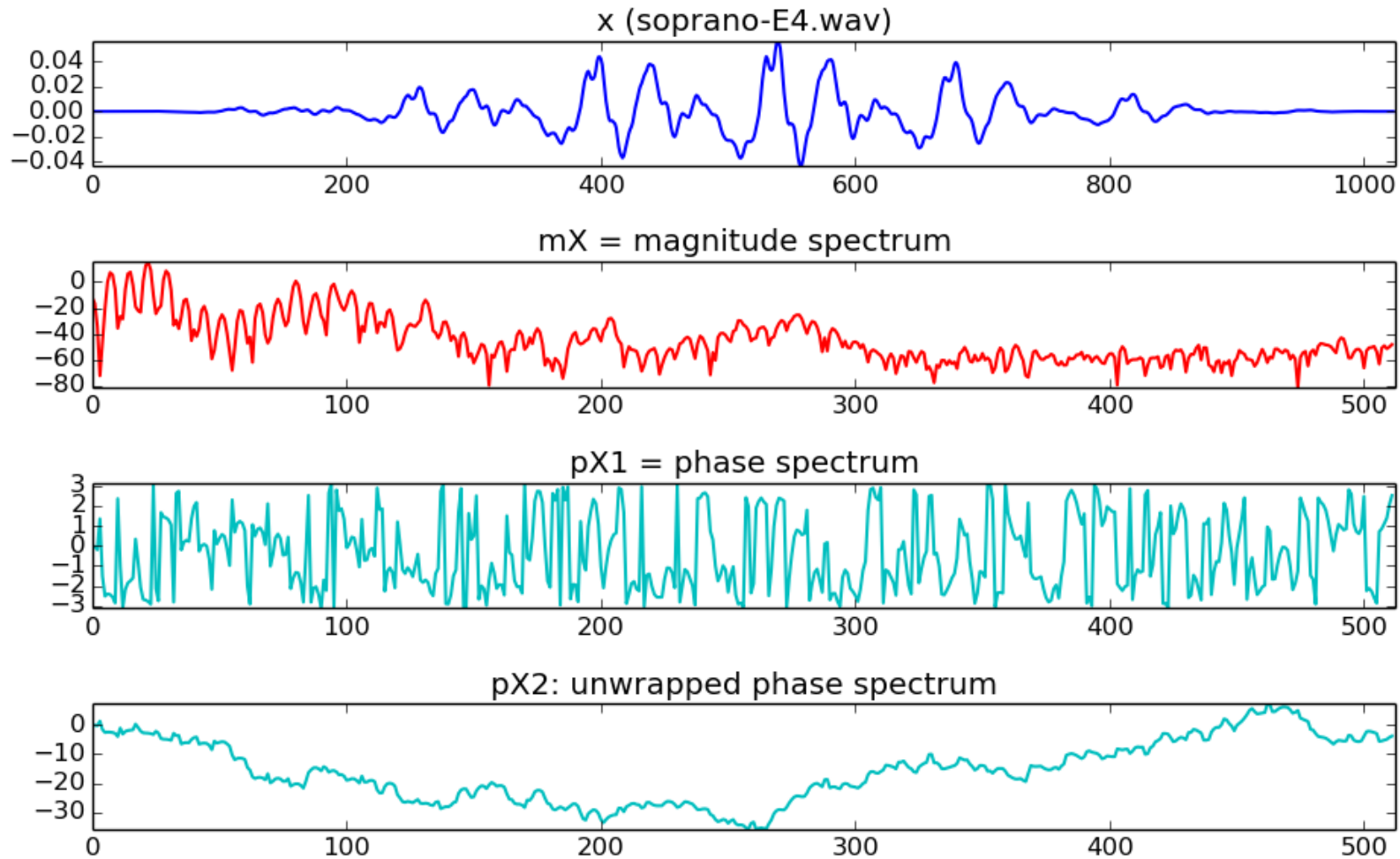
$$X_{\text{dB}}[k] = 20 \log_{10}(\max(|X[k]|, \epsilon) / \text{ref})$$



## dB scale

- $20 \log_{10}$  maps amplitude ratios to dB.
- Small magnitudes become visible.
- Use a floor before log to avoid  $-\infty$ .

# Phase unwrapping



## Phase

- Wrapped phase jumps at  $\pm\pi$ .
- Unwrapping reveals continuous trends.
- Shifts produce phase ramps.

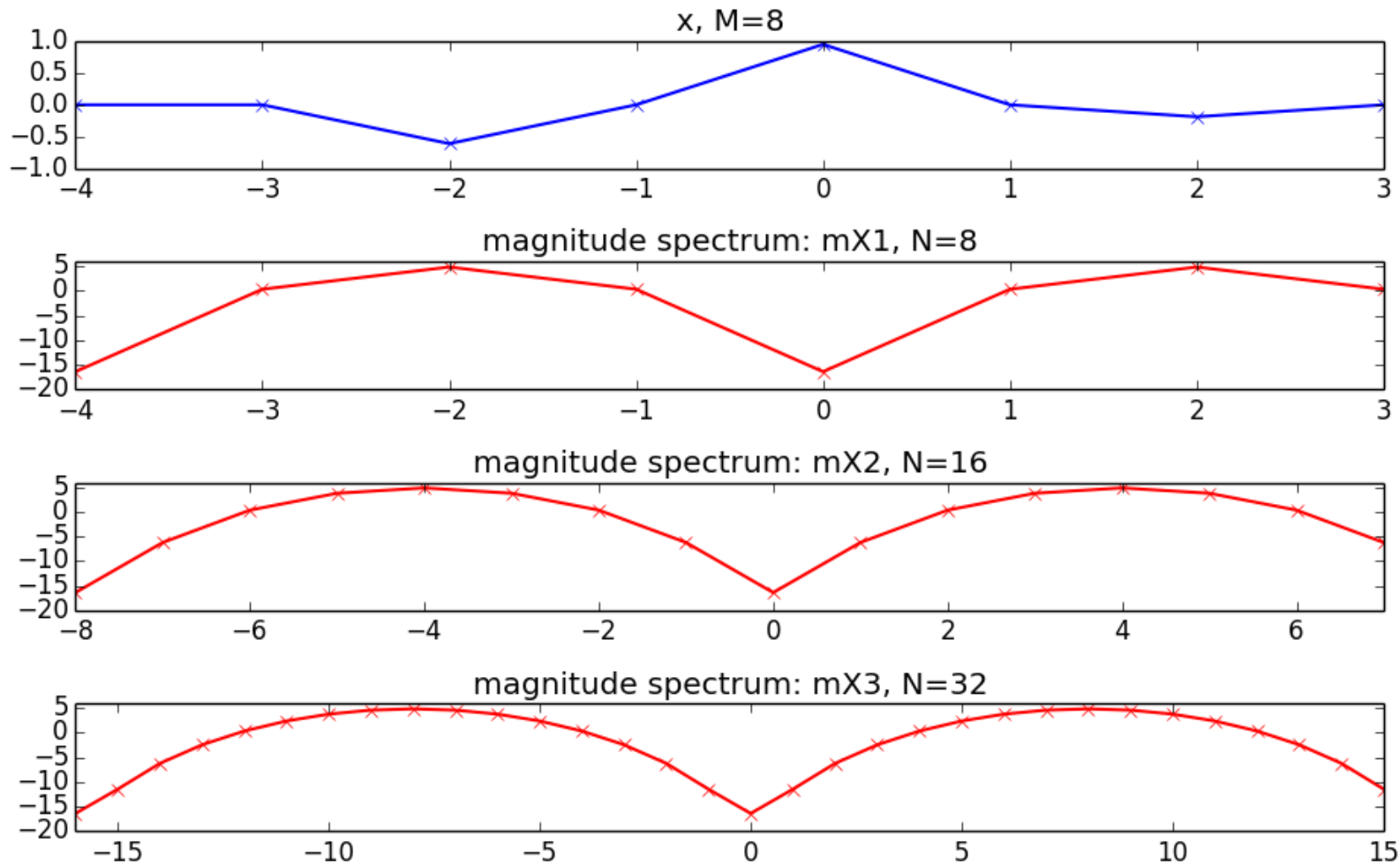
# Zero-padding

zero-padding  $\Leftrightarrow$  interpolation

$$x_{zp}[n] = x[n] \text{ for } 0 \leq n < M; 0 \text{ otherwise}$$

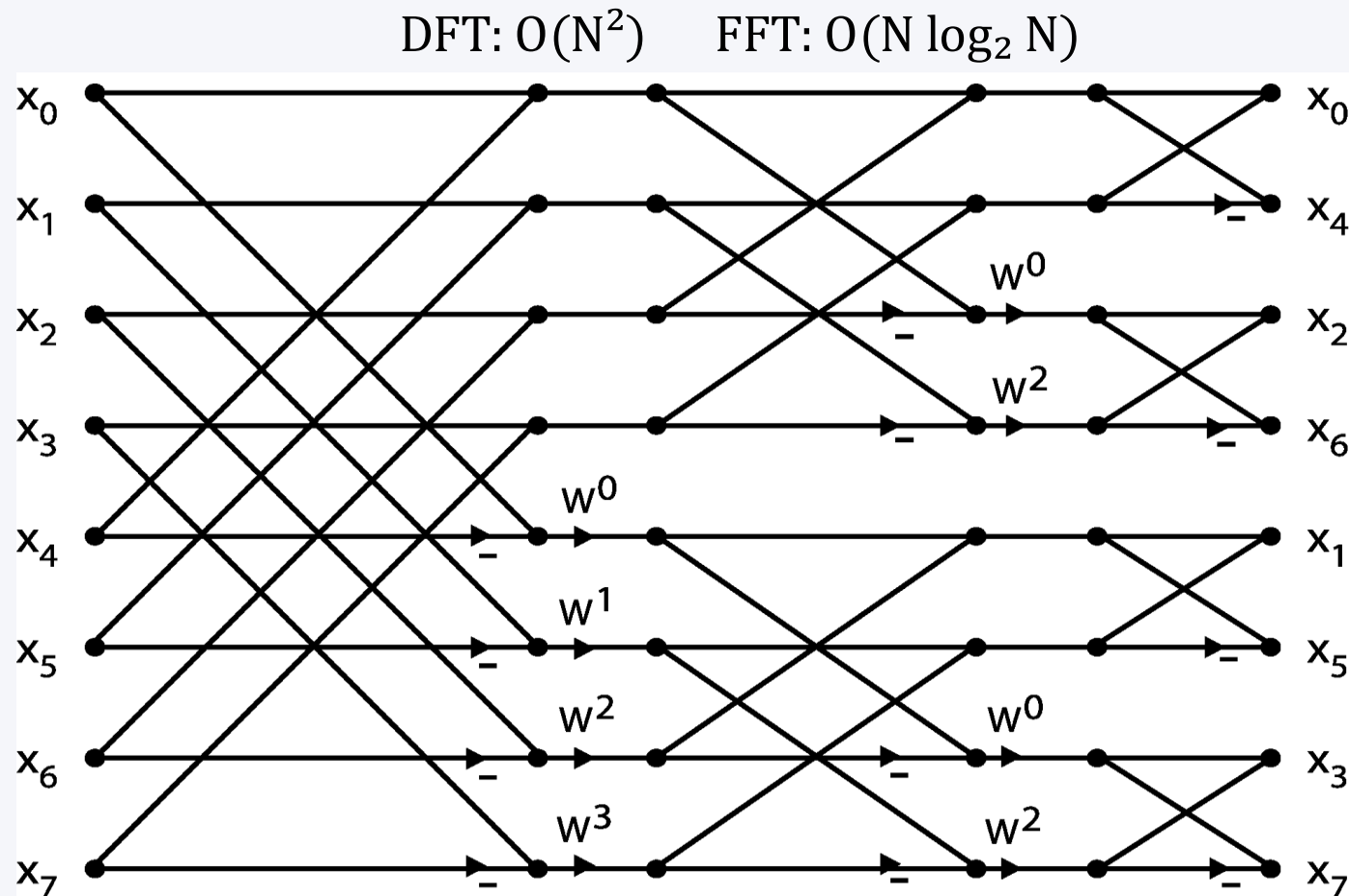
## Zero-padding

- Does not add information.
- Interpolates the DFT samples.
- Helps visualize and locate peaks.



# Fast Fourier Transform

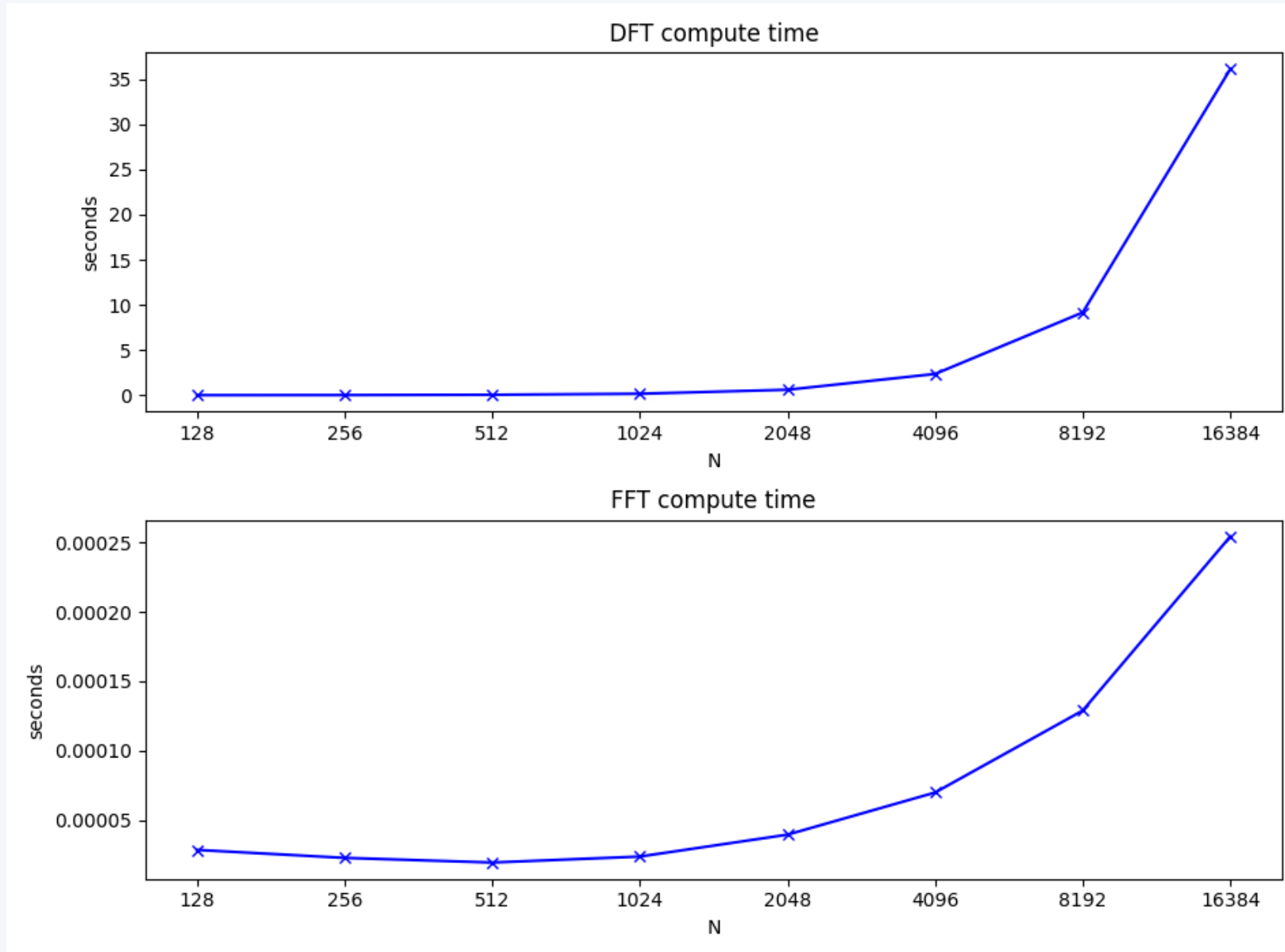
*Cooley-Tukey algorithm:* breaks down recursively the DFT of a power of 2 size into two pieces of size  $N/2$ .



## FFT

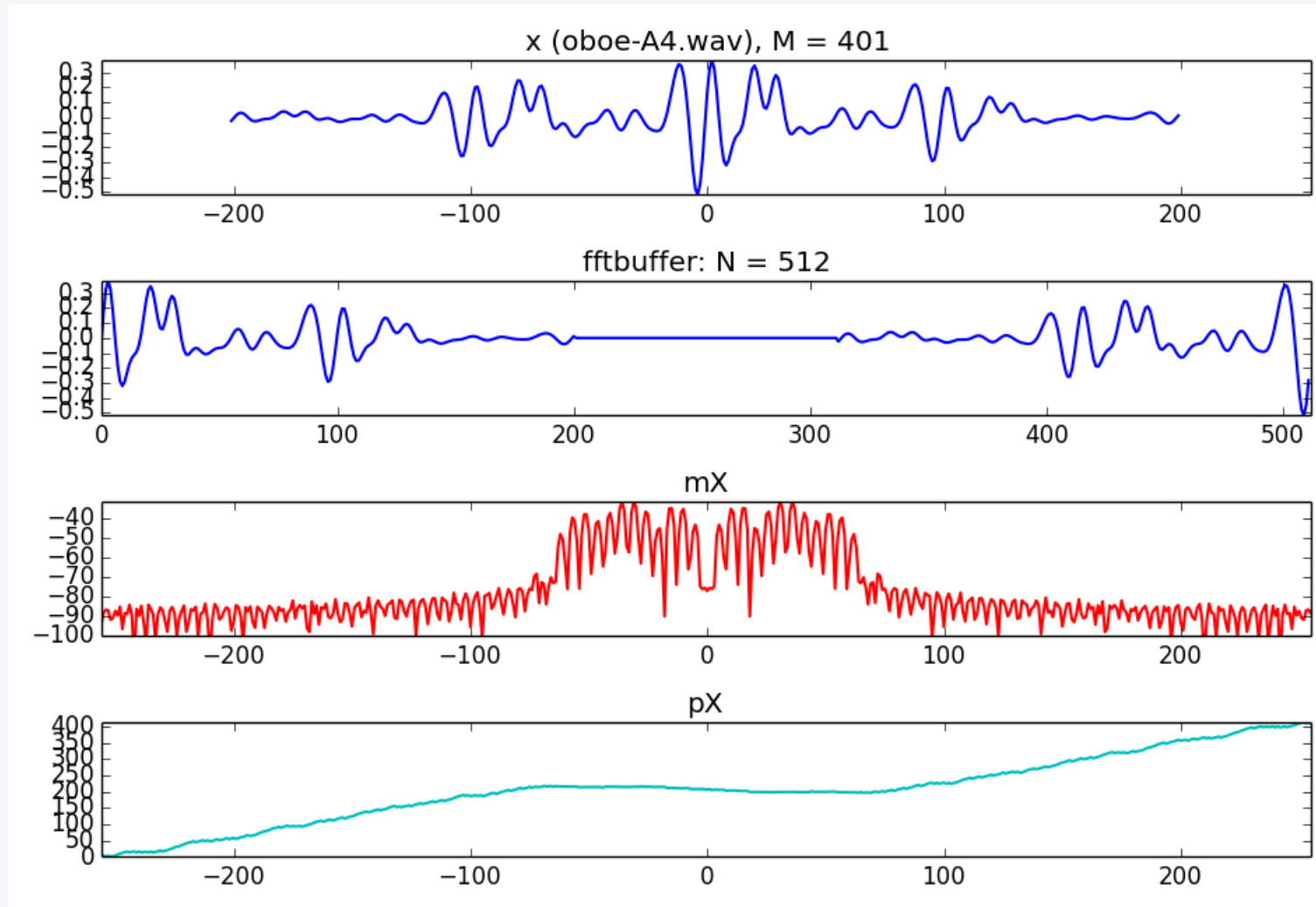
- FFT computes the same DFT.
- It reduces cost from  $O(N^2)$  to  $O(N \log N)$ .
- Power-of-two sizes are often efficient.

# DFT vs FFT computational cost



# FFT and zero-phase windowing

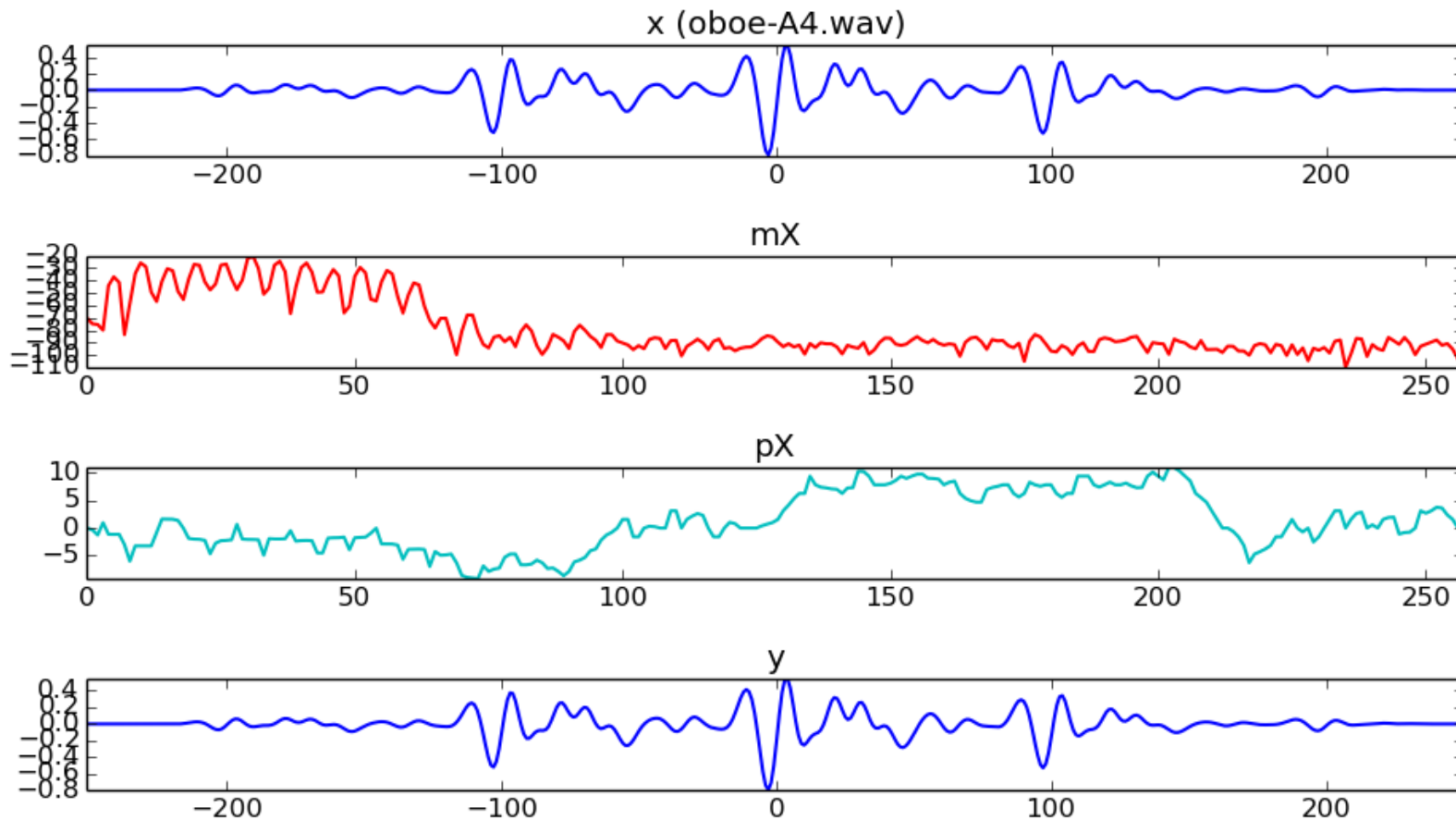
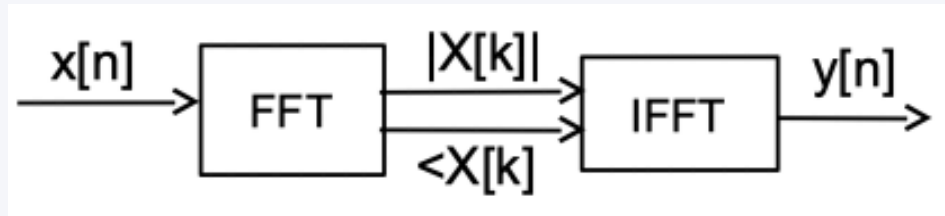
zero-phase FFT: center the frame around  $n=0$  before computing  $X[k]$



## Zero phase

- Center the frame around  $n = 0$ .
- Reorder samples into the FFT buffer.
- Avoid an artificial phase ramp.

# Analysis/synthesis with magnitude and phase



## Analysis / synthesis

- FFT decomposes the frame.
- Magnitude and phase preserve information.
- IFFT reconstructs the signal.

# Example notebooks: Fourier properties in practice

Use `examples/dft-examples.ipynb` to verify that every spectrum is tied to a precise time-domain operation: framing, windowing, centering, FFT, and inverse FFT.

## 1 • Frame and center

Select one short sound frame and place it around  $n = 0$  when phase interpretation matters. A non-centered frame introduces a phase ramp.

## 2 • Window the frame

Apply a Hann/Hamming-type window to reduce boundary discontinuities. Then the spectral shape belongs to the windowed signal, not the raw frame.

## 3 • Inspect spectrum

Look at magnitude and phase together. For real signals, use conjugate symmetry as a consistency check: magnitudes mirror, phases oppose.

## 4 • Reconstruct

Use the full complex spectrum for IFFT. Zero-padding can make plots smoother, but it does not add information.

# How to read the example outputs

## Windowing

Compare the spectrum before and after windowing. A smoother window reduces discontinuities at the frame boundaries and controls spectral leakage.

## Symmetry

For real-valued frames, check that positive and negative frequency bins are conjugate pairs. Magnitudes mirror; phases have opposite signs.

## Zero-phase / phase

Centering the frame around  $n = 0$  removes the artificial phase ramp caused by a time shift. This makes phase easier to interpret.

# What to remember from Topic 3

- Fourier properties explain how time-domain operations affect spectra.
- Centering and windowing are essential for interpretable phase.
- Zero-padding interpolates; it does not increase resolution.
- FFT is an efficient DFT implementation.
- Analysis/synthesis requires both magnitude and phase.

# Exercise 3: Fourier properties

Third lab: turn Fourier properties into small experiments, plots, checks, and listening tests.

## What you will practice

- predict how time operations affect spectra
- compare magnitude, phase, real, and imaginary parts
- use windowing, zero-phase placement, FFT, and IFFT
- separate spectral leakage from zero-padding interpolation

## How to start

**Open** `exercises/E3-Fourier-properties.ipynb`

Use: NumPy · Matplotlib · DFT/FFT · IFFT · plots

Compare with: `examples/dft-examples.ipynb`

## Lab structure

### 1 · Leakage

- choose M carefully
- compare matched / mismatched frames

### 2 · Symmetry

- real, even, odd signals
- check expected DFT structure

### 3 · Filtering

- edit spectral bins
- synthesize and listen

### 4 · FFT size

- compare M and N
- zero-padding interpolates

### 5 · Verify

- magnitude + phase
- numerical check + comment

## What you will implement in Exercise 3

- Choose  $M$  to minimize spectral leakage for two sinusoids using the LCM of periods.
- Test real/even/odd symmetry using zero-phase windowing and inspect real/imaginary parts.
- Manipulate spectral components in the DFT domain and synthesize the filtered signal.
- Compare window size  $M$  and FFT size  $N$  to separate resolution from zero-padding interpolation.
- Always verify with both magnitude and phase, not magnitude alone.

# References and credits

- Original MTG sms-tools-materials lecture slides and plotting code.
- plots-code: [lectures/03-Fourier-properties/plots-code](#)
- Example notebook: [examples/dft-examples.ipynb](#)
- Exercise notebook: [exercises/E3-Fourier-properties.ipynb](#)
- Julius O. Smith, Mathematics of the Discrete Fourier Transform.
- Slides follow the CC Attribution-NonCommercial-ShareAlike spirit of the original materials.